Enco_Deco: Project Specification Report

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GitHub: https://github.com/Mandeep-pro/Enco_Deco

1 Overview

Enco_Deco is an interactive C++17 application that visualizes digital line coding and delta modulation.

- **Digital mode:** random, constrained-zero-run(length 4 or 8), or manually entered bit sequences; longest palindromic substring via Manacher's algorithm, $O(n)$; NRZ-L, NRZ-I, Manchester, Differential Manchester and AMI with optional B8ZS/HDB3 scrambling of AMI.
- **Analog mode:** configurable sine-wave sampling, one-bit delta modulation with staircase reconstruction, waveform plots, and absolute-error metrics.

2 Language, Libraries and Tools

- **Language:** C++17
- **Graphics/UI:** Raylib (windowing, interface and plotting)
- **Build/Test:** CMake (configuration and build), CTest (automated tests)
- **Version Control:** Git / GitHub

3 Assumptions

- **Signal levels:** the positive level is $+1$, the negative level is -1, and the zero level is 0.
- **NRZ-L:** 1 maps to $+1$ and 0 maps to -1.
- **NRZ-I:** a 0 keeps the current level and a 1 inverts it. The initial level is $+1$, so the first 1 gives -1.
- **Manchester (IEEE 802.3):** a 1 is a low-to-high transition $(-1,+1)$ and a 0 is a high-to-low transition $(+1,-1)$.
- **Differential Manchester:** there is always a mid-bit transition, and a 0 also has a transition at the start of the bit. The initial level is $+1$.
- **AMI:** a 0 maps to 0 and each 1 alternates polarity. The previous pulse polarity starts at -1, so the first 1 is $+1$.
- **B8ZS/HDB3:** applied only on top of AMI, using the standard substitution rules for eight and four consecutive zeros respectively.
- **Analog input:** a sampled sine wave with positive parameters and $f < f_s/2$. Delta modulation uses a fixed one-bit up/down step.

4 Build, Run and Test

Install a C++17 compiler, CMake and the Raylib development files, then run from the repository root:

- **Configure:** `cmake -S . -B build`
- **Build:** `cmake -build build -parallel`
- **Run:** `./build/bin/enco_deco`
- **Test:** `ctest -test-dir build -output-on-failure`

5 References

- Course lecture slides from Piazza
- CP-Algorithms, *Manacher's algorithm*: `https://cp-algorithms.com/string/manacher.html`
- Raylib documentation: `https://www.raylib.com/cheatsheet/cheatsheet.html`
- LearnCpp: `https://www.learncpp.com`

6 Acknowledgment

Generative AI tools(ChatGPT and Claude) were used as supplementary resources for assisting GUI development, resolving technical issues, and refining the build and testing workflow. The authors reviewed and integrated the resulting changes into the project.